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Final Project

Recommender Systems in Social Media Advertising

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Abstract

Recommender systems play an increasingly important role in shaping the digital information environment. On social media platforms in particular, algorithmic systems determine which content and advertisements are shown to which users. These systems are typically optimized for engagement metrics such as clicks or viewing time. While this can increase advertising efficiency and user engagement, it may also introduce biases and influence how information and opportunities are distributed across different user groups.

This project investigates how different advertisement recommender system designs influence both recommendation performance and the distribution of exposure across users. We compare two recommendation strategies using a social media engagement dataset. The first system is a popularity-based recommender that ranks advertisements according to overall engagement across all users. The second system is a personalized recommender that predicts the probability that a specific user will engage with a specific advertisement using both user characteristics and advertisement features.

The systems are evaluated using common recommender system metrics such as Precision@K and Recall@K. In addition to predictive performance, the study analyses how the different algorithms distribute advertisement exposure across demographic groups. By combining a technical evaluation with theoretical concepts such as data bias, system bias, and algorithmic fairness, this project highlights how design decisions in recommender systems can shape both algorithmic outcomes and social consequences.

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1 Introduction

Recommender systems have become a central component of modern digital platforms. These systems are widely used to personalize content and guide user attention across large online environments. On social media platforms, recommender systems play an especially important role in determining which advertisements users encounter. By analyzing large amounts of user interaction data, platforms attempt to optimize the distribution of advertisements in order to maximize engagement and advertising revenue.

This development is closely connected to the process of datafication (Strikovic 2025c). In a digital society, many aspects of everyday life are transformed into data that can be analyzed and used by algorithmic systems. Activities such as browsing, liking, sharing, or clicking generate behavioral data that platforms can use to infer user preferences and predict future interactions Mejias and Couldry (2019). As a result, recommender systems rely heavily on data that is generated through everyday user behavior.

However, the increasing reliance on data-driven systems raises important societal and ethical questions (Milano, Taddeo, and Floridi 2020). Data is not neutral, and the way in which it is collected and processed can influence algorithmic outcomes. Decisions about what data is collected, how it is structured, and how it is used in algorithms can shape the behavior of recommender systems and their impact on users. These design choices may advantage certain stakeholders while disadvantaging others.

One important concern is algorithmic bias. When recommender systems are trained on historical user data, they may reproduce patterns that already exist in the dataset. For example, if certain groups of users interact more frequently with particular types of advertisements, recommendation algorithms may learn to target these advertisements toward those groups. This can lead to unequal exposure of content and reinforce existing patterns in the data (Zuiderveen Borgesius et al. 2016).

Another important issue is related to algorithmic gatekeeping. Traditionally, the selection and distribution of information in media environments were controlled by human editors and journalists. In digital environments, these decisions are increasingly made by algorithms that automatically prioritize and rank content. Recommender systems therefore play a role in shaping what information becomes visible to users and what remains unseen.

In this project, we investigate how different recommender system designs influence both recommendation performance and the distribution of advertisement exposure. Specifically, we compare a popularity-based recommender system with a personalized recommender system. The popularity-based system recommends advertisements that have historically received the highest engagement across all users. In contrast, the personalized system predicts the probability that a specific user will interact with a specific advertisement.

By comparing these two approaches, this project examines how algorithmic design choices influence both technical performance and broader societal outcomes. In particular, we analyze whether different recommendation strategies lead to differences in advertisement exposure across demographic groups and whether popularity-based recommendation systems amplify already dominant advertisements. Based on these considerations, this study addresses the following research question:

How do different recommender system designs (popularity-based vs. personalized) affect both recommendation performance and the distribution of social media advertisement exposure across demographic groups?

2 Theoretical Framework

Recommender systems are not purely technical systems. They operate within broader social and economic environments and influence how information and opportunities are distributed across users. As a result, evaluating recommender systems requires both technical analysis and theoretical reflection on their societal implications.

2.1 Stakeholders

Recommender systems affect multiple stakeholders simultaneously. First, users interact with the system and receive recommendations based on their behavior and inferred preferences. Their experience is directly influenced by the relevance, diversity, and fairness of the recommendations they receive (Brajnik and Gabrielli 2010).

Second, advertisers depend on recommender systems to reach potential audiences. The visibility of their advertisements depends on how recommendation algorithms rank and distribute advertising content across users.

Third, platform providers design and operate the recommender systems. These platforms have strong economic incentives to optimize algorithms for engagement and revenue generation.

Finally, society itself can also be considered a stakeholder. Because recommender systems influence how information and commercial messages are distributed at large scale, they can shape the broader digital information environment (Royne Stafford and Pounders 2021). Decisions about algorithmic design therefore have implications that extend beyond individual users or advertisers.

2.2 Biases and Fairness

Bias in recommender systems can emerge at multiple stages of the system design process. One source of bias is the underlying data used to train the algorithm. If certain groups of users are overrepresented or under-represented in the dataset, the resulting models may produce biased outcomes. This phenomenon is often referred to as data bias (Strikovic 2025a).

Bias can also emerge from the design of the recommendation algorithm itself. For example, recommender systems that prioritize engagement may exhibit popularity bias. In such systems, items that already receive high engagement are recommended more frequently, which further increases their visibility. This feedback loop can lead to unequal exposure of content and marginalize less popular items (Strikovic 2025a).

Another related concept is exposure bias. Because recommender systems influence which items users interact with, they also shape the data that will be used for future recommendations. This creates a feedback cycle in which algorithmic decisions influence future training data (Strikovic 2025a).

The concept of fairness in recommender systems is complex and often involves trade-offs between different goals (Strikovic 2025b). Systems may aim to maximize accuracy and engagement while also attempting to ensure fair exposure of content or equal treatment of users. However, improving fairness along one dimension may reduce performance along another dimension. Designers of recommender systems must therefore make explicit decisions about which outcomes to prioritize.

In this project, fairness is examined by analyzing how different recommender system designs distribute advertisement exposure across demographic groups. By comparing a popularity-based recommender with a personalized recommender, we aim to better understand how algorithmic design choices influence both recommendation performance and potential biases in the distribution of advertisements.

3 Method

3.1 Dataset

This study uses a social media advertisement engagement dataset that records user interactions with online advertisements (Ziya 2025). The dataset contains information about users, advertisement characteristics, and engagement-related metrics. Each observation represents a user–advertisement interaction and includes attributes describing both the user and the advertisement.

User-related variables include demographic characteristics such as age, gender, and geographic location. Advertisement-related variables describe properties of the advertisement itself, including the advertisement type, the advertisement category, and the duration of the advertisement in seconds. Furthermore, the dataset includes behavioural attributes such as the number of previous interactions with similar advertisements, the device used to view the advertisement, and the time of day at which the advertisement was viewed.

The dataset also contains several engagement-related variables, including the number of clicks, likes, shares, and the time spent viewing the advertisement. The target variable is *engaged*, a binary variable that shows whether the user interacted with the advertisement. This variable is used as the main outcome for training and evaluating the recommender systems.

The choice of dataset is fundamental to the research method, as it determines which user and advertisement attributes are available for analysis and therefore shapes how the recommender systems can be built. In this study, the dataset provides both demographic user information and advertisement characteristics, which enables the comparison between a popularity-based recommendation strategy and a personalized recommendation approach.

3.2 Data Preprocessing

Before constructing the recommender systems, the dataset was preprocessed to ensure it could be effectively used for model training and evaluation. Missing or incomplete observations were removed where necessary to maintain data consistency.

A data schema was constructed, where appropriate user and advertisement features were selected from the original dataset. These features for users include age, gender, geographic location and device type. The selected advertisement features include type, category, duration and finally a binary engagement metric. The construction of both recommender systems in this study rely on these selected features.

To evaluate the performance of the recommender systems, the dataset was divided into a training set and a test set. The training set was used to construct the recommendation models, while the test set was used to evaluate their predictive performance on unseen user–advertisement interactions.

3.3 Recommender Systems

Two different recommender systems were implemented and compared in this study: a popularity-based recommender and a personalized recommender.

The popularity-based recommender ranks advertisements based on their overall engagement across all users in the training dataset. Advertisements with higher levels of engagement are considered more popular and are therefore recommended more frequently. This approach assumes that advertisements that have historically generated high engagement are more likely to produce engagement in the future.

The personalized recommender system takes into account both user characteristics and advertisement features. In this approach, a predictive model is trained to estimate the probability that a specific user will engage with a specific advertisement. The model uses the earlier mentioned user and advertisement features. Based on these predicted engagement probabilities, advertisements with the highest predicted scores are recommended to each user.

By comparing these two systems, the study evaluates the difference between a general popularity-based recommendation strategy and a more personalized recommendation approach.

3.4 Evaluation

The recommender systems were evaluated using both performance-based and distribution-based metrics. To measure recommendation performance, common recommender system metrics were used, including Precision@K and Recall@K. These metrics evaluate how well each system recommends advertisements that users actually engage with.

Precision@K measures the proportion of recommended advertisements among the top K results that resulted in user engagement, while Recall@K measures the proportion of all relevant advertisements that were successfully recommended within the top K results.

In addition to predictive performance, the study also evaluates fairness and bias of the recommender systems. Firstly, we evaluate exposure distributions differences between demographic groups using Jensen-Shannon divergence. In addition, accuracy across these groups is tested using gap ration and epsilon-squared effect. Furthermore, fairness is evaluated using the Herfindahl-Hirschman index. Bias amplification is evaluated by comparing each group's engagement in training data against their recommendation exposure. Intersectional analysis examines disadvantage at gender-times-age-group intersections. Finally, we use Maximal Marginal Relevance, which balances relevance with category diversity via a tunable parameter lambda.

4 Results

4.1 Predictive performance

The predictive performances, measured by precision and recall of the built recommender systems, are shown in table 1. These results indicate that the popularity based recommender outperformed the personalised recommender on both precision and recall for small K-values. However, as K-values increase, the personalised recommender catches up in predictive performance and eventually outperforms the popularity recommender on both metrics. This catch up in performance seems to happen earlier for recall, indicating that the popularity based recommender will accurately make generalized recommendations, but quickly start missing out on recommendations for specific users.

Metric	Popularity	Personalized
Precision@5	0.0152	0.0136
Recall@5	0.0337	0.0329
Precision@10	0.0144	0.0138
Recall@10	0.0629	0.0674
Precision@20	0.0126	0.0132
Recall@20	0.1187	0.1299

Table 1: Predictive performance comparison.

4.2 Distributive performance

The distributive performances, measured by a diverse set of metrics, are shown in table 2. The popularity recommender scores 0.0 on exposure divergence for all tested demographic groups, indicating that every group is shown the same advertisements. The popularity recommender, however shows some exposure divergence for gender, but large divergence for age, meaning age groups see very different ads.

The accuracy gap ratio between genders is more significant for the personalized recommender, showing a higher precision for men. The popularity recommender however, performs more evenly across different gender groups.

Regarding provider fairness, both recommender scores similarly on HHI, indicating that a select few of providers dominate the recommendations. However, the personalized recommender has a significantly higher catalogue coverage, indicating a more diverse arrangement of ads being show.

Furthermore, the personalized recommender shows severe bias amplification. Conversely, the popularity recommender doesn't seem to show any significant stereotyping patterns.

Finally, the popularity recommender manages to pass 5 out of 8 fairness criteria set, failing primarily on catalogue coverage. On the other hand, the personalized recommender only passes 2 criteria, mainly showing issues in exposure divergence, accuracy equity and bias amplification.

Metric	Popularity	Personalized
Exposure JSD (gender)	0.00	0.14
Exposure JSD (age, max)	0.00	0.77
Accuracy gap ratio (gender)	0.70	0.36
HHI	0.20	0.27
Catalogue coverage	5.0%	24.0%
Max stereotyping amplification	+15pp	+55pp
Fairness score	5/8 PASS	2/8 PASS

Table 2: Distributive performance comparison.

5 Discussion

5.1 Interpretation of Results

The results show a clear difference between the two recommendation strategies. The popularity-based recommender performs better at lower values of K, indicating that it is effective at identifying generally engaging advertisements. However, its performance does not scale as well when more recommendations are considered.

In contrast, the personalized recommender performs better at higher values of K, suggesting that it is more effective at capturing individual user preferences when a broader set of recommendations is evaluated.

5.2 Biases and fairness

A key finding of this study is the trade-off between recommendation performance and fairness. While the personalized recommender achieves higher predictive performance at larger values of K, it also introduces several concerns regarding bias and fairness. In particular, exposure divergence and heavy stereotyping occur.

The popularity-based recommender, on the other hand, produces a more even distribution of advertisements across users, resulting in lower exposure divergence. However, this comes at the cost of reduced personalization and lower performance for larger recommendation sets. More prominently, the reinforcement of popularity bias, is a strong concern of implementing such a system.

These findings highlight a core trade-off when building recommender systems: optimizing for individual relevance may lead to unequal outcomes, but better predictive performance, while more general recommendation strategies may reduce bias but limit personalization as well as predictive performance.

5.3 Limitations

This study has several limitations. First, the dataset represents a set moment of user interactions and does not capture dynamic behaviour that happens in real-world recommender systems. In practice, recommendations further influence future user behaviour, which in turn affects future recommendations.

Second, the recommenders are based on a limited set of user and advertisement features dependent on the chosen dataset. Additional variables, may improve performance and alter fairness outcomes.

Finally, the evaluation focuses on a specific set of fairness metrics. Different definitions of fairness may lead to different conclusions regarding the relative performance of the recommender systems.

5.4 Implications and recommendations

One potential improvement is finding a compromise between different recommender system implementations. Finding such a balance could potentially optimize performance, while also maintaining fairness and avoiding bias.

In addition, recommender systems could be built and adjusted based on fairness outcomes. By identifying weaknesses and repeatedly readjusting, a system could be gradually improved.

Finally, increasing transparency in recommendation algorithms may help stakeholders better understand and evaluate the trade-offs involved in different system designs.

6 Conclusion

This study set out to answer the following research question: How do different recommender system designs (popularity-based vs. personalized) affect both recommendation performance and the distribution of social media advertisement exposure across demographic groups?

The results show that the two recommendation strategies show clear differences in both performance and distributional outcomes. The popularity-based recommender performs better at lower values of K and provides a more even distribution of advertisements across users. However, it reinforces popularity bias and offers limited personalization. In contrast, the personalized recommender performs better predictively at higher values of K and is more effective at identifying individual user preferences, but it introduces substantial exposure divergence and amplifies existing biases in the data.

These findings demonstrate that recommender system design involves a crucial trade-off between accuracy and fairness. Systems that optimize for personalization may improve predictive performance, but risk losing fairness and introducing more bias, while more general recommendation strategies may reduce disparities, but at the cost of personalization and predictive performance.

Overall, this study highlights the importance of considering both technical performance and societal implications when designing recommender systems, particularly regarding advertising. Future work should focus on developing approaches that find a suitable balance between these points, for example by implementing fairness-aware mechanisms or hybrid strategies that combine personalization with fairer exposure.

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